

YACHIKA ANAND

AI Engineer | Generative AI & RAG Systems | Azure Data & AI Solutions

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PROFESSIONAL SUMMARY

- AI and data professional with 6+ years of experience delivering GenAI, RAG, ML, analytics, and governed data solutions across enterprise, education, healthcare, and product environments.
- Build deployment-ready AI applications using LangGraph, LangChain, Azure OpenAI, FastAPI, vector search, evaluation workflows, streaming interfaces, and responsible AI controls.
- Translate business workflows into secure AI/data solutions, executive dashboards, and measurable delivery plans that improve knowledge access, decision quality, and operational confidence.

TECHNICAL SKILLS

- Generative AI & LLMs:** LangChain, LangGraph, LangSmith, Azure OpenAI, OpenAI API, RAG, agentic workflows, embeddings, semantic search, hybrid retrieval, prompt engineering, guardrails, AI evaluation
- Application Engineering:** Python, FastAPI, Streamlit, Pydantic, REST APIs, async programming, Server-Sent Events, Docker, GitHub Actions
- Data & Cloud Platforms:** Microsoft Azure, Azure AI Foundry, Azure Data Factory, Azure SQL, Blob Storage, Azure ML, Microsoft Fabric, Databricks, Synapse Analytics, Spark, Delta Lake, Kafka, Snowflake
- Databases & Vector Stores:** PostgreSQL, Redis, Qdrant, SQL, vector search, semantic caching
- ML, Analytics & BI:** Scikit-learn, PyTorch, TensorFlow, Hugging Face, XGBoost, Pandas, NumPy, Power BI, DAX, Tableau, Plotly
- MLOps & Governance:** MLflow, DVC, Great Expectations, Evidently AI, model monitoring, structured logging, data lineage, responsible AI,

PROFESSIONAL EXPERIENCE

Data Solutions Architect & AI Lead | NWLB | USA (Remote) | Jan 2026 - Present

- Lead Azure AI and data platform delivery across GenAI/RAG, governed data pipelines, and executive reporting, connecting stakeholder requirements to technical architecture and delivery plans.
- Build RAG and agentic workflow patterns using LangChain, LangGraph, Azure OpenAI, governed enterprise knowledge sources, retrieval design, prompt controls, and evaluation criteria.
- Delivered an Azure data pipeline with automated validation, lineage, and governance checkpoints to produce analytics-ready datasets for reporting, dashboarding, and downstream AI consumption.
- Launched an executive Power BI reporting suite with automated refresh, DAX KPI models, and data quality rules, reducing manual reporting effort and improving stakeholder confidence.
- Established practical AI delivery controls including access controls, prompt documentation, evaluation metrics, responsible AI notes, and observable implementation practices.
- Partner with business stakeholders to define AI use cases, success criteria, risk controls, and adoption steps before deployment.

Professor | Conestoga College | Canada | Jan 2023 - Apr 2025

- Guided 100+ capstone teams through ML, analytics, and data-product lifecycles, coaching on problem framing, data validation, feature engineering, model evaluation, deployment concepts, and stakeholder communication.
- Reviewed ML and analytics projects for metric integrity, data leakage risk, validation design, interpretability, reproducibility, responsible use, and business relevance.
- Designed applied labs in Python, SQL, Power BI, D3.js, ETL workflows, cloud analytics, statistics, machine learning, and business intelligence for 200+ learners annually.
- Mentored learners on turning raw data into model-ready datasets, dashboards, experiments, and stakeholder-ready technical narratives.

Analyst & Web Developer | IEC Hamilton | Canada | Aug 2023 - Apr 2024

- Built a Power BI analytics dashboard integrating multi-source marketing data, DAX measures, KPI definitions, and validation controls to improve decision visibility.
- Conducted A/B testing, funnel analysis, and statistical evaluation that supported a 25% conversion-rate increase through evidence-based improvements.

Professor - Data Analytics | Anovus Institute of Clinical Research | India | Oct 2019 - Dec 2022

- Applied statistical analysis, data governance, validation, and reproducibility practices in a regulated clinical research environment using SAS, Excel, and Python.
- Developed workshops on data manipulation, model-ready datasets, audit-ready documentation, regulatory data practices, and quality controls aligned with clinical research standards.

Data Analysis Trainer | Sai Institute of Skill Development | India | 2018 - 2019

- Delivered hands-on SQL, Python, and Excel training to 120+ learners using project-based exercises based on real business datasets, supporting a 96% graduate placement rate.

SELECTED PROJECTS

ResearchHub - AI-Powered Research Assistant Platform

GitHub: <https://github.com/yachika-yashu/Research-hub>

Tech: Python, FastAPI, LangGraph, Qdrant, Redis, PostgreSQL, Streamlit, Docker

- Built a research productivity platform enabling semantic search, AI-driven Q&A, multi-paper comparison, md note-taking, and usage tracking over uploaded papers.
- Implemented hybrid retrieval combining dense vectors, BM25 sparse retrieval, Reciprocal Rank Fusion, and reranking for higher-precision results.
- Designed Redis exact-match caching and Qdrant semantic similarity caching to reduce repeated LLM calls, API cost, and response latency.
- Integrated LangGraph for stateful agent workflows with persistent chat history, structured logging, and usage governance.

MedVault - Agentic RAG Research Platform

GitHub: <https://github.com/yachika-yashu/MedVault>

Tech: Python, FastAPI, LangGraph, LangChain, Qdrant, Redis, PostgreSQL, Streamlit, Docling, OpenAI API

- Designed a full-stack agentic RAG platform where an LLM agent can search a private vault, query arXiv, and auto-ingest papers
- Engineered a multimodal PDF ingestion pipeline using Docling to preserve table and figure references, chunk documents, and store hybrid dense and sparse embeddings.
- Implemented team-scoped multi-tenancy with tenant IDs enforced at the vector query layer, strengthening security by design.
- Delivered a real-time streaming UI with SSE that surfaces agent tool calls, ingestion progress, reasoning state, and LLM token output.

Production AI Performance Benchmarking Toolkit

GitHub: <https://github.com/yachika-yashu/production-ai-performance-benchmarking-toolkit.git>

Tech: Python, OpenAI API, LangChain, RAGAS-style evaluation, Pandas, Qdrant, FastAPI, Streamlit, Pytest

- Built a production-grade benchmarking toolkit for LLM and RAG systems covering latency profiling, quality scoring, retrieval optimization, cost analysis, and regression testing.
- Implemented RAGAS-style metrics for faithfulness, answer relevance, context precision, and context recall to compare prompts, retrieval strategies, chunking approaches, and model configurations.
- Created prompt A/B testing workflows that log prompts, contexts, generated answers, token usage, estimated cost, latency, and evaluation scores for repeatable benchmark runs.

Clinical Trial Data Pipeline - Real-Time ETL for Healthcare

GitHub: <https://github.com/yachika-yashu/ClinETL-Pipeline>

Tech: Kafka, Spark Structured Streaming, PostgreSQL, Delta Lake, GE, Evidently AI, XGBoost, Isolation Forest, Docker, Streamlit, FastAPI

- Designed a real-time ETL pipeline for 500-patient Phase 3 clinical trial data, processing approximately 10K records daily with Spark Structured Streaming.
- Implemented FDA 21 CFR Part 11 and HIPAA-aligned controls including immutable audit trails, AES-256 PII encryption, data lineage, and schema validation and Built 40+ GE validation rules with bad-record handling and data quality monitoring, achieving a 95%+ quality score.

Databricks Retail Lakehouse - Medallion Data Platform

GitHub: <https://github.com/yachika-yashu/databricks-retail-lakehouse.git>

Tech: Databricks, Delta Lake, PySpark, Spark SQL, Medallion Architecture, Power BI, CSV, JSON, Parquet

- Designed and implemented an end-to-end data lakehouse using Databricks and medallion architecture, reducing data processing time from 72 hours to under 6 hours and improving analytics accuracy by 25%.
- Built a scalable ETL pipeline processing 10B+ annual transactions across Bronze, Silver, and Gold layers with ACID transactions using Delta Lake and PySpark.
- Integrated Power BI dashboards for real-time analytics and reporting across 27 countries and 11K+ retail locations.

EDUCATION

- M.Eng., Information Technology | Punjab University, India | 2019 - 2021
- B.Eng., Electronics & Communication Engineering | DAV Institute, India | 2013 - 2017

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Microsoft Certified: Azure Fundamentals | Microsoft | Mar 2026
- Cloud Computing, Big Data & AI - Bridge to IT Careers | Seneca Polytechnic | Jan - Apr 2026
- Secure Coding - Career Acceleration Program | University of Waterloo, Watspeed | Nov 2024 - Jan 2025
- In Progress: Microsoft Azure AI Engineer Associate; Azure Data Engineer Associate

PUBLICATIONS

- Yachika, Kaur, P., & Garg, R. (2021). A survey on key enabling technologies towards 5G. IOP Conference Series: Materials Science and Engineering, 1033, 012011.
- Energy Efficient Resource Allocation using Tabu Search and Gradient Descent Method in 5G Wireless Devices. Journal of Xidian University, 15(8), 2021.